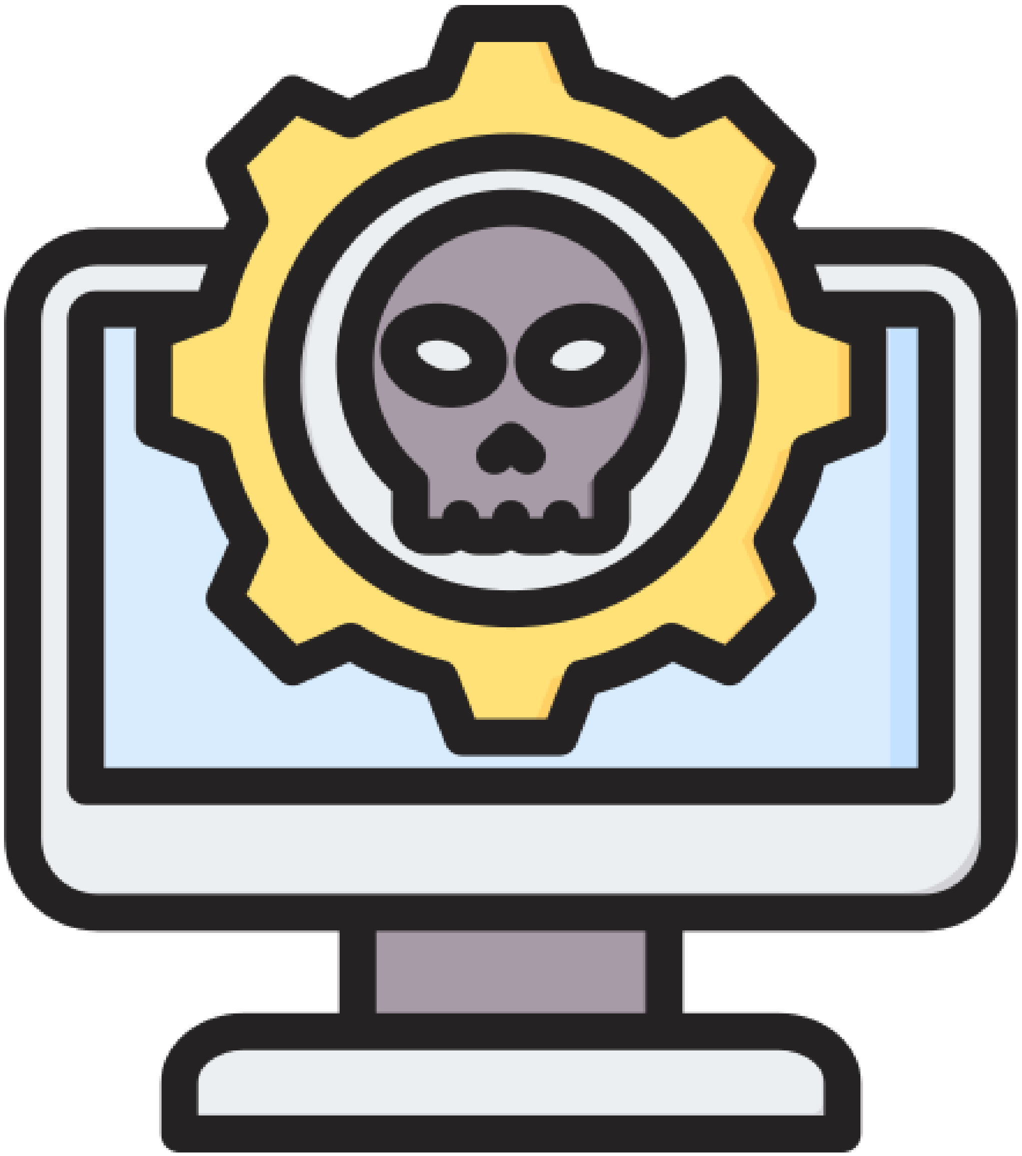
ET 8202: INTRODUCTION TO IS SECURITY

LECTURE 10

Saturday, January 13, 2024

Malicious Software



What is Malware?

- Software intended to intercept or take partial control of a computer's operation without the user's informed consent.
- Piece of software designed with intent of compromising the security of another software/system.
- **Malware covers all kinds of intruder software**
  - ✓ Including viruses, worms, backdoors, rootkits, Trojan horses, stealware etc. These terms have more specific meanings.

What is Malware?

- **The Purpose of Malware:** To partially control the user's computer, for reasons such as;
  - ✓ To subject the user to advertising,
  - ✓ To launch DDoS on another service,
  - ✓ To track the user's activity ("spyware"),
  - ✓ To spread spam,
  - ✓ To commit fraud, such as identity theft and affiliate fraud, . . . and perhaps other reasons

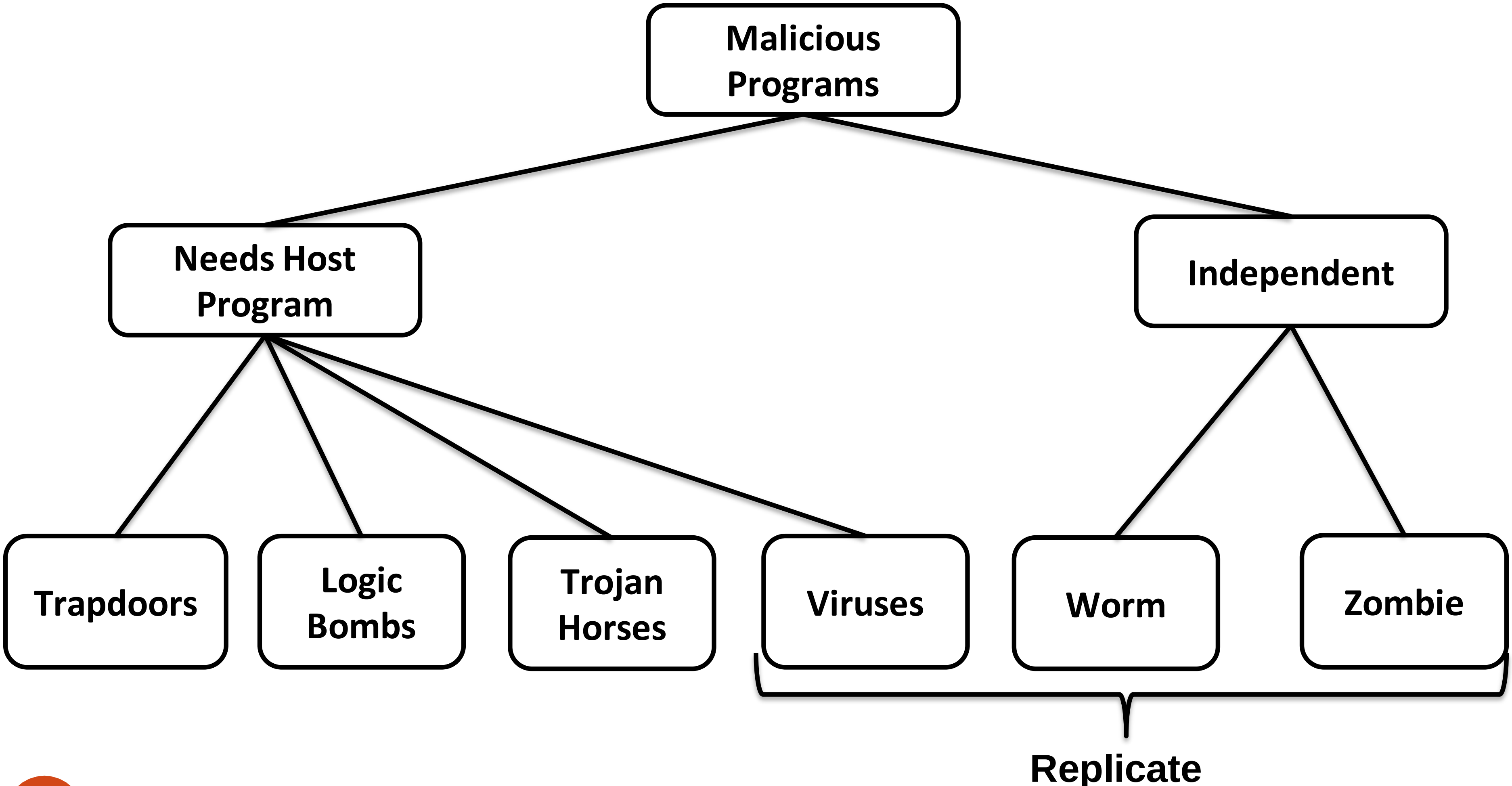
Malicious Programs

Two categories

- **Those that need a host program:** Fragments of programs that cannot exist independently of some application program, utility, or system program.
- **Those that are independent:** Self-contained programs that can be scheduled and run by the operating system (self contained)

Malicious Programs

Taxonomy of Malicious Programs



Malicious Programs

- **Logic Bombs (**also called **slag code):** Logic embedded is a program that checks for a set of conditions to arise (such as the lapse of a certain amount of time or the failure of a program user to respond to a program command) and executes some function resulting in unauthorized actions.
- **Trapdoors:** Secret undocumented entry point into a program, used to grant access without normal methods of access authentication.
- **Trojan Horse:** Secret undocumented routine embedded within a useful program, execution of the program results in execution of the routine. Common motivation is data destruction.

Malicious Programs

- **Zombie:** A program that secretly takes over an Internet attached computer and then uses it to launch an untraceable attack. Very common in **Distributed Denial-Of-Service** attacks
- **Spyware:** The term "spyware" taken literally suggests software that surreptitiously monitors the user.
- **Virus:** A virus is a self-replicating program that produces its own code by attaching copies of itself into other executable codes.

Malicious Programs

- Some viruses affect computers as their code is executed; others viruses lie dormant until a pre- determined logical circumstance is met.
- **Characteristics of Virus:**
  - Infects other program
  - Transforms itself,
  - Encrypts itself,
  - Alter data,
  - Corrupts files and programs
  - Self propagation.

Malicious Programs

Virus Life Time

- **Dormant**: The Virus is idle and waiting
- **Propagation**: The Virus place a copy of itself into other programs
- **Trigger**: Virus is activated to perform function for which it was intended
- **Execution**: Virus function is performed

Malicious Programs

- ***Four stages of virus lifetime***

1. **Dormant phase:** Here, the **virus remains idle** and gets activated based on a certain action or event(for example, a user pressing a key or on a certain date and time etc)
2. **Propagation phase:** The virus starts propagating, that is multiplying itself (**cloning of virus**). A piece of code copies itself and each copy starts copying more copies of self, thus propagating.
3. **Triggering phase:** A Dormant virus moves into this phase when **it gets activated,** that is, the event it was waiting for gets initialized.
4. **Execution phase:** This is **the actual work of the virus**. It can be destructive(deleting files on disk) or harmless(popping messages on screen).

Others: Stages of virus life (6 Stages)

1. **Design stage:** developing virus code using programming languages or construction kits.
2. **Replication stage:** virus replicate for a period of time within the target system and then spreads itself
3. **Launch stage:** it gets activated with the user performing certain actions such as running an affected program.
4. **Detection stage:** a virus is identifies as threat infection target systems
5. **Incorporation stage:** ant-virus software developers assimilate defenses against the virus.
6. **Elimination stage:** users install ant-virus updates and eliminate the virus threats

Computer Virus

- **Avoiding Detection: Infected version** of program is **longer** than the corresponding uninfected one
  - ***Solution****:* **compress** the **executable file** so infected and uninfected versions are identical in length

- **Encryption in the operation of a virus:** A portion of the virus, generally called a *mutation engine*, creates a random encryption key to encrypt the remainder of the virus.
  - The key is stored with the virus, and the mutation engine itself is altered. When an infected program is invoked, the virus uses the stored random key to decrypt the virus.
  - When the virus replicates, a different random key is selected.

Computer Virus

- **Why do people create computer virus?**
  - Inflict damage to competitors
  - Vandalism
  - Financial benefits
  - Cyber terrorism
  - Research projects
  - Distributed political massages
- **How does a computer get infected by virus?**
  - Not running the latest ant-virus application
  - Not updating and not installing new versions of plug-ins
  - Installing pirated software
  - Opening infected e-mail attachments
  - When a user accepts files and downloads without checking properly for the source.

Indications of virus attack

- **Abnormal activities:** Is the systems acts in unprecedented manner, you can suspect a virus attack.
  - ✓Processes take more resources and time
  - ✓Computer beeps with on display
  - ✓Driver label changes
  - ✓Unable to load Operating System
  - ✓Anti-virus alerts
  - ✓Browser window “freezes”
  - ✓Hard drive is accessed often
  - ✓Files and folders are missing
  - ✓Computer freezes frequently or encounters errors
  - ✓Computer slows down when programs start.
- Note: false positive
  - ✓However, not all glitches can be attributed to virus attacks

Types of viruses

- System or boot sector viruses
- Files virus
- Cluster viruses
- Macro virus
- Multipartite virus

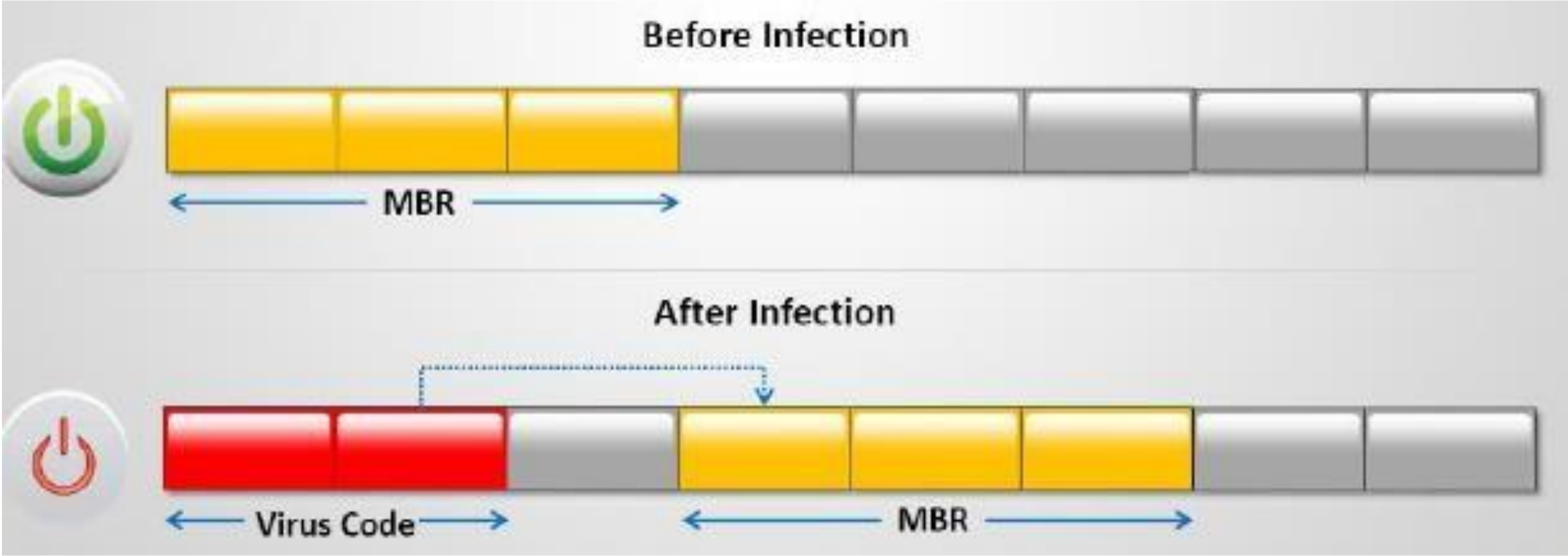
What do they infect?

How do they infect?

- Stealth virus/tunneling virus
- Encryption virus
- Polymorphic virus
- Overwriting file or cavity virus
- Sparse virus
- Companion virus/camouflage virus
- Shell virus
- File extension virus
- Add-on virus
- Intrusive virus
- Direct action or transient virus
- Terminate and stay resident virus (TSR)

System or boot sector viruses

- Boot sector virus moves **master boot record (**MBR) to another location on the hard disk and copies itself to the original location of MBR
- When system boots, virus code is executed first and then control is passed to original MBR



MBR: Master Boot Record

Types of viruses…

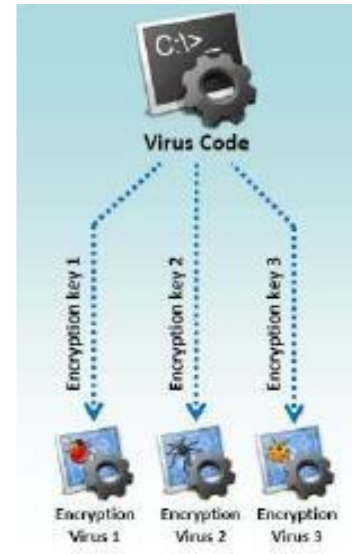
- **Files and multipartite viruses**
  - **File virus:** File virus infect files which are **executed** or interpreted is the system such as COM, EXE, SYS, OVL, OBJ, PRG, MNU and BAT files
  - **Multipartite virus:** Multipartite virus that attempts to attack both the **boot sector** and the **executable** or **program files** at the same time
- **Macro virus**
  - Macro viruses infect files created by Microsoft Word or Excel
  - Most macro viruses are written using macro language Visual Basic for Application (VBA)
  - Macro viruses infect templates or convert infected documents into template files, while maintaining their appearance if ordinary document files.

Types of viruses…

- **Cluster viruses**
  - Cluster viruses **modify directory table entries** so that directory entries point to the virus code instead of the actual program.
  - There is only copy of the virus on the disk infecting all the programs in the computer system
  - It will launch itself first when any program on the computer system is started and then the control is passed to actual program
- **Stealth/tunneling viruses**
  - These viruses evade the anti-virus software by **intercepting** its requests to the operation system
  - A virus can hide itself by intercepting the anti-virus software's request to read the file and passing the request to the virus, instead of the OS.
  - The virus can then **return an uninfected version of the file** to the ant-virus software, so that it appears as if the file is "clean"

Types of viruses…

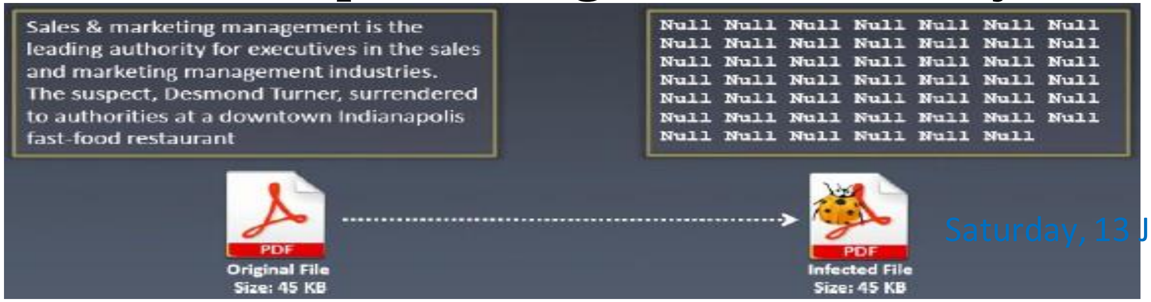
- **Encryption viruses**
  - ✓ This type of virus uses simple encryption to encipher the code
  - ✓ The virus is encrypted with a different key for each infected files
  - ✓ **AntiVirus** (AV) scanner cannot directly detect these types of viruses using **signature detection** method.



- **Polymorphic code**
  - ✓ Polymorphic code is a code that mutates (**constantly changes**) while keeping the original algorithm intact
  - ✓ To enable polymorphic code, the virus has to have a polymorphic engine (also called mutation engine or mutation engine)
  - ✓ A well-written polymorphic virus therefore has no parts that stay same on each infection making it **difficult to detect** with anti-malware programs.

Types of viruses…

- **Metamorphic viruses**
  - Metamorphic viruses rewrite themselves completely each time they are to infect new executable.
  - Metamorphic code can reprogram itself by translating its own code into a **temporary representation** and then back to the normal code again.
  - For example, W32/simile consisted of over 1400 lines of assembly code, 90% lines of assembly code, 90% of it is of the metamorphic engine.
- **File overwriting or cavity viruses:** Cavity virus **overwrites** a part of the host file with a constant (usually nulls), without increasing the length of the file and preserving its functionality.



Types of viruses…

❖ **Sparse Infector viruses**

- ✓Sparse infector virus infects only **occasionally** (e.g. every 10th program execute), or only files whose lengths falls within a narrow range.
- ✓By infection less often, such viruses try to minimize the probability of being discovered.

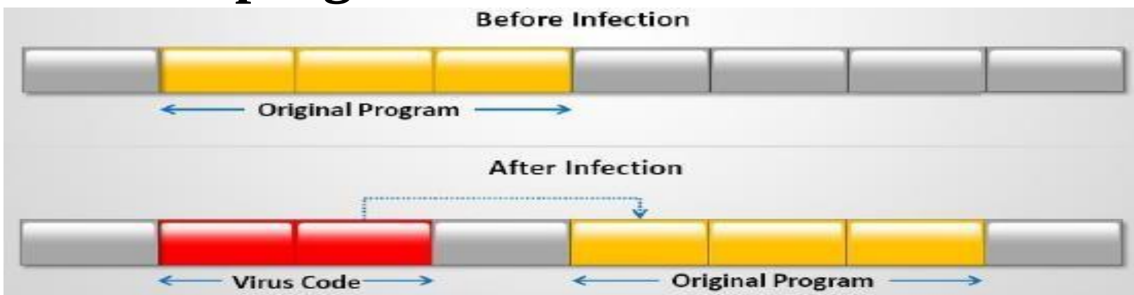
Types of viruses…

❖ Companion/camouflage viruses

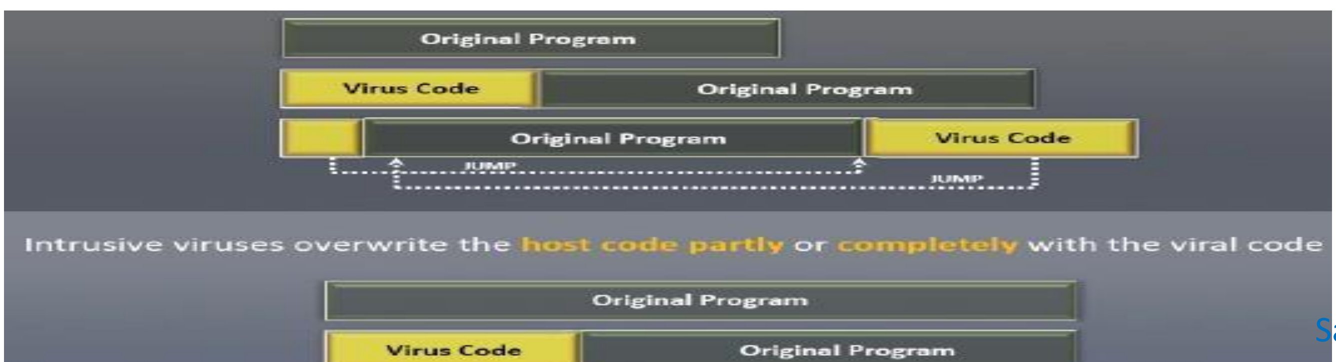
- ✓ A companion virus **creates a companion file** for each executable file the virus infects.
- ✓ Therefore, a companion virus may save itself as notepad.com and every time a user executes a notepad.exe (good program), the computer will load notepad.com (virus) and infect the system.

Types of viruses…

- **Shell viruses**
  - ✓ Virus code forms a shell around the target host program's code, making **itself the original program** and **host code as its sub-routine**.
  - ✓ Almost **all boot program viruses are shell viruses**



- **Add-on and intrusive viruses:** Add-on viruses **append** their code to the host code without making any changes to the latter or relocate the host code to insert their own code at the beginning.



File extension viruses

- File extension viruses **change the extensions of the files**
- .TXT is safe as it indicates as pure text file
- With extension turned off, if someone sends you a file named BAD.TXT.VBS, you will only see BAD.TXT
- If you have forgotten that extensions are turned off, you might think is a text file and open it.
- This is an executable visual basic script virus file and could do serious damage.
- Countermeasure is turn off “Hide file extensions” in windows.

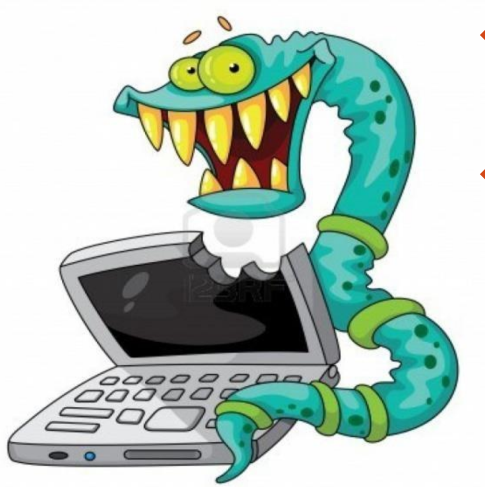
Transient and terminate and stay resident viruses

Basic infection techniques

- **Direct action or transient virus**
  - ✓ Transfers all the controls of the host code to where it resides
  - ✓ Selects the target program to be modified and corrupt it
- **Terminate and stay resident virus (TSR)**
  - ✓ Remains permanently in the memory during the entire work session even after the target host's program is executed and terminated; can be removed only by rebooting the system.

Computer worms

- Computer worms are malicious programs that replicate, execute, and spread across the network connections indecently without human interaction.
- Most of the worms are created only to replicate and spread across a network, consuming available computing resources; however some worms carry a payload to damage the host system.
- Attackers use worm payload to install backdoors in infected computers, which turns them into zombies and creates botnet; these botnets can be used to carry further cyber attacks.



- Unlike a computer virus, it does not need to attach itself to an existing program.
- Worms almost always cause at least some harm to the network, even if only by consuming bandwidth, whereas viruses almost always corrupt or modify files on a targeted computer.

How is worm different from a virus?

- A worm is special type of malicious software that can replicate itself and use memory, but cannot attach itself to other programs.
- A worm takes advantage of file information transport features on computer systems and spreads through the infected network automatically but a virus does not.

Virus detection methods

- **Scanning**
  - Once a virus has been detected, it is possible to write scanning programs that look for **signature** string characteristics of the virus.
- **Integrity checking**
  - Integrity checking products work by reading the entire disk and recording integrity data that acts as a **signature** for the files and system sectors.
- **Interception**
  - The interceptor monitors the operation system requests that are written to the disk.

Viruses Countermeasures

1. Antivirus approaches
2. Advanced antivirus techniques
   - ✓ Generic Decryption
   - ✓ Digital Immune System
3. Behavior-blocking software

Viruses Countermeasures

1. Antivirus Approaches

- **Detection :** Determine that it has occurred and locate the virus
- **Identification:** Identify the specific virus
- **Removal :** Remove all traces and restore the program to its original state

❖ Generations of Antivirus Software

- ✓ **First:** Simple scanners (record of program lengths)
- ✓ **Second:** Heuristic scanners (integrity checking with checksums)
- ✓ **Third:** Activity traps (memory resident, detect infected actions)
- ✓ **Fourth:** Full-featured protection (suite of antivirus techniques, access control capability)

WHAT ARE THE CURRENT AVAILABLE ANTIVIRUS PROGRAMS?

Company	Windows	Apple	Linux	Mobile	Free?
AntiVir	Yes	No	Yes	No	Yes
AVG	Yes	No	No	No	Yes
Avira	Yes	No	Yes	Yes	Yes
BitDefender	Yes	No	Yes	Yes	No
ClamWin	Yes	No	No	No	Yes
ESET NOD32	Yes	No	Yes	Yes	No
F-Prot	Yes	No	Yes	No	No
Kaspersky	Yes	Yes	Yes	Yes	No
McAfee	Yes	Yes	Yes	Yes	No
MSE	Yes	No	No	No	Yes
Network Associates	Yes	Yes	Yes	Yes	No
Panda Software	Yes	No	Yes	No	No
RAV	Yes	Yes	Yes	No	No
Sophos	Yes	Yes	Yes	No	No
Symantec (Noton)	Yes	Yes	Yes	Yes	No
Trend Micro	Yes	No	No	Yes	No
Vipre	Yes	No	No	No	No
Webroot	Yes	No	No	No	No

Viruses Countermeasures

2. Advanced Antivirus Techniques

- Generic Decryption
- Digital Immune System

Generic Decryption

- Generic decryption (GD) technology enables the antivirus program to **easily detect** even the **most complex** polymorphic viruses and other malware, while maintaining fast scanning speeds.
- Contains following elements:
  - ✓ **CPU emulator:** software based virtual computer. Instructions in an executable file are interpreted by the emulator rather than executed on the underlying processor so that the underlying processor is unaffected by programs interpreted on the emulator.
  - ✓ **Virus signature scanner:** scans target code looking for known signatures
  - ✓ **Emulation control module:** control execution of target code. Thus, if the code includes a decryption routine that decrypts and hence exposes the malware, that code is interpreted. In effect, the malware does the work for the anti-virus program by exposing itself. Periodically, the control module interrupts interpretation to scan the target code for malware signatures.

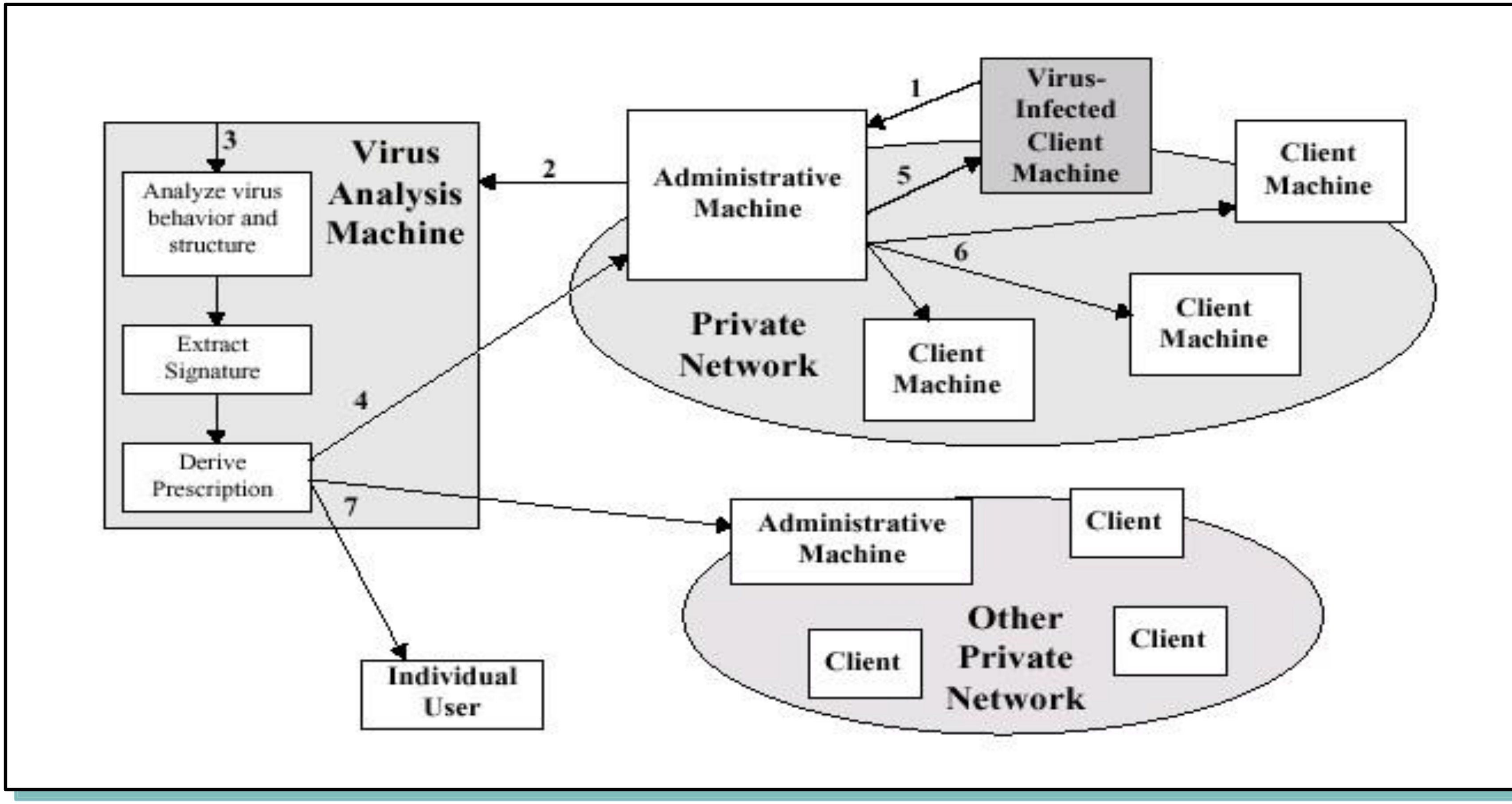
Digital Immune System

- **DIS (Digital Immune System):** A closed-loop, suspect-code submission system designed to detect unknown but potentially malicious code, quarantine the code, submit it for analysis, and finally push out new virus definitions to affected systems.
- Developed by IBM (refined by Symantec) for general purpose emulation and virus detection system
- **Motivation:** rising threat of internet-based virus propagation
  - ✓Integrate mail systems (eg MS outlook)
  - ✓Mobile-program system (eg Java and ActiveX)
- Expands the **use** of program **emulation**
- Depends on a **central Virus Analysis Machines (VAM)**

Digital Immune System

- This system provides a general-purpose emulation and **virus-detection** system. The objective is to provide **rapid response time** so that viruses can be stamped out almost as soon as they are introduced.
- When a new virus enters an organization, the immune system automatically captures it, analyzes it, adds detection and shielding for it, removes it, and passes information about that virus to systems running a general antivirus program so that it can be **detected before** it is allowed to run elsewhere.

Digital Immune System



1. Each PC runs a monitoring program to detect unusual behavior
2. Encrypt the sample and forward to VAM
3. Analyze the sample in a safe environment via emulation
4. Prescription via sent back to Admin. Machine

5-6 Forwarded to the infected client as well as the other PCs on the same network

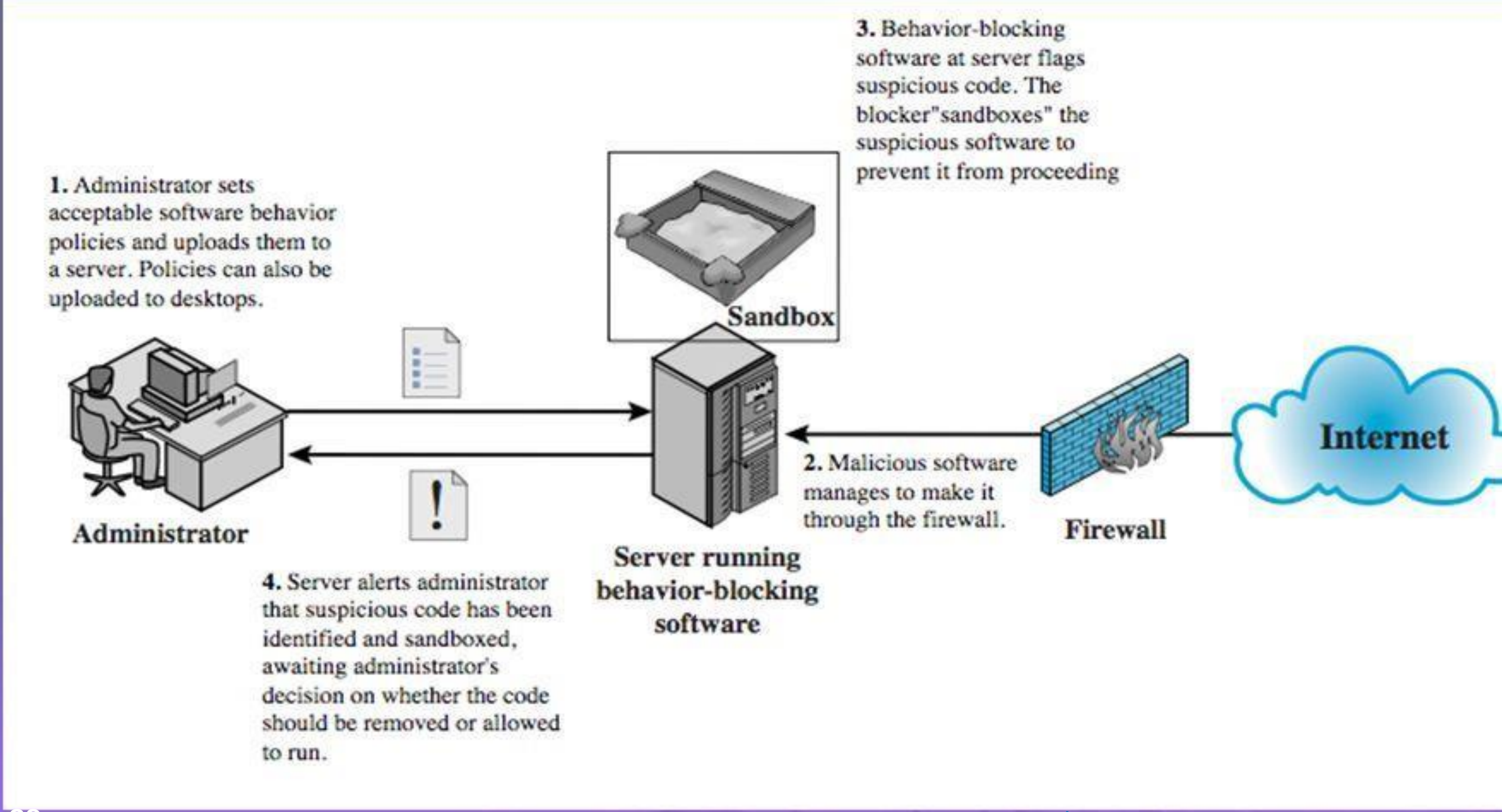
7. All subscribers receive regular antivirus updates

Viruses Countermeasures

3. Behavior-blocking Software

- Behavior-blocking software integrates with the operating system of a host computer and monitors program behavior in real time for malicious actions.
- The behavior blocking software then blocks potentially malicious actions before they have a chance to affect the system.
- IPS – **Intrusion Prevention Systems**
- Monitored behaviors can include
  - ✓ Attempts to open, view, delete, and/or modify files;
  - ✓ Attempts to format disk drives and other unrecoverable disk operations;
  - ✓ Modifications to the logic of executable files or macros;
  - ✓ Modification of critical system settings, such as start-up settings;
  - ✓ Scripting of e-mail and instant messaging clients to send executable content;
  - ✓ Initiation of network communications.

Behavior-Blocking Software



Malicious Code Protection
Types of Products

- **Scanners** - identify known malicious code - search for ***signature strings***
- **Integrity Checkers** – determine if code has been altered or changed – ***checksum*** based
- **Vulnerability Monitors** - prevent modification or access to particularly sensitive parts of the system – user defined
- **Behavior Blockers** - list of rules that a legitimate program must follow – ***sandbox*** concept

Virus and worms countermeasures (others)

- Ensure the executable code sent to the organization is approved
- Do not boot the machine with infected bootable system disk
- Know about the latest virus threats
- Check the DVDs and CDs for virus infection
- Ensure the pop-up blocker is returned on use an internet firewall
- Run disk clean up, registry scanner and defragmentation once a week
- Block the files with more than one file type extension
- Be caution with the files being sent through the internet messenger.

Virus and worms countermeasures (Others)

- Install ant-virus software that detects and removes infections as they appear
- Generate an anti-virus policy for safe computing and distribute it to the staff
- Pay attention to instructions while downloading files or any programs from the Internet
- Update the ant-virus software on the monthly basis, so that it can identify and clean out new bugs

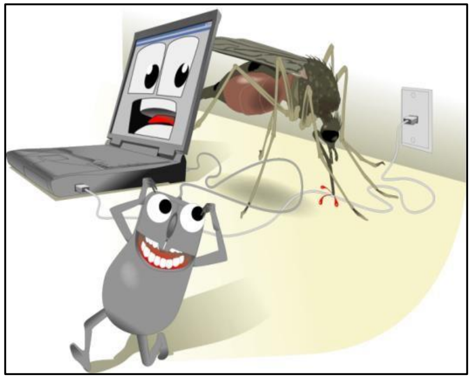
Virus and worms countermeasures (others)

- Avoid opening the attachments received form an unknown sender as virus spread via e-mail
- Possibility of virus infection may corrupt data, thus regularly maintain data back up
- Schedule regular scans for all drivers after the installation of ant-virus
- Do not accept disks or programs without checking them first using a current version of anti-virus program.

PRACTICE:
USE CARE WHEN READING EMAIL WITH ATTACHMENTS

- Executable content
- Interesting to you (social engineering)
- Violates trust
- **KRESV tests**
  - ✓ **Know test:** Know the sender?
  - ✓ **Received test:** Received email before?
  - ✓ **Expect test:** Did you expect this email?
  - ✓ **Sense test:** Does this email make sense?
  - ✓ **Virus test:** Contain a virus?
- Doesn't pass all tests? Don't open!
- **Level of effort: High**

INSTALL AND USE ANTIVIRUS SOFTWARE

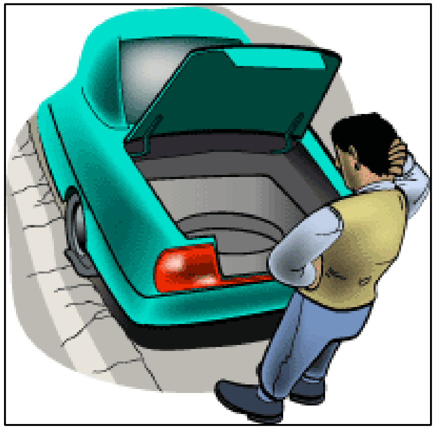


- Easy way to gain control of your computer or account
- Violates "trust"
- **DURCH tests**
  - **Demand**: Check files on demand?
  - **Update**: Get new virus signatures automatically?
  - **Respond**: What can be done to infected files?
  - **Check**: Test every file for viruses.
  - **Heuristics**: Does it look like a virus?
- **Level of effort: low**

PRACTICE:

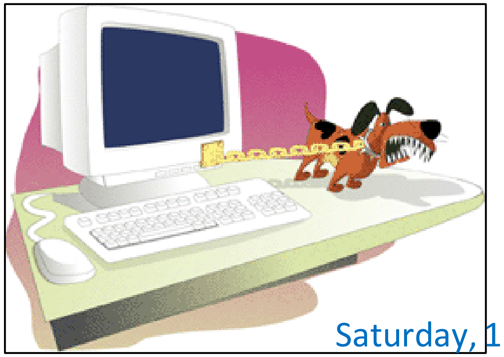
MAKE BACKUPS OF IMPORTANT FILES AND FOLDERS

- Can you recover a file or folder if lost?
- Does your computer have a “spare tire”?
- **FOMS tests**
  - **Files**: What files should be backed up?
  - **Often**: How often should a backup be made?
  - **Media**: hat kind of media should be used?
  - **Store**: Where should that media be stored?
- Level of effort:
  - **setup: medium to high**
  - **maintaining: medium**



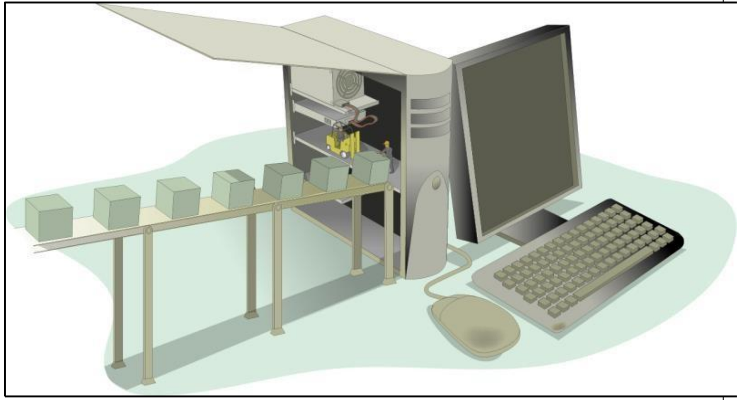
INSTALL AND USE A FIREWALL PROGRAM

- Limit connections to computer
- Limit connections from computer based on application
- Portable – follows the computer (laptop)
- **PLAT tests**
  - **Program** – What program wants to connect?
  - **Location** – Where does it want to connect?
  - **Allowed** – Yes or no?
  - **Temporary** – Permanent or temporary?
- Level of effort:
  - **install: low**
  - **maintain: high**

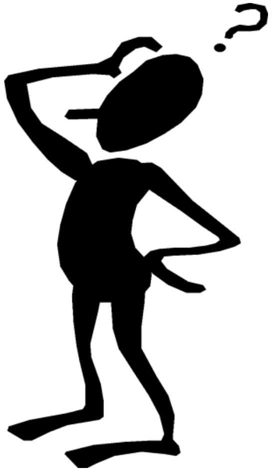


USE CARE WHEN DOWNLOADING AND INSTALLING PROGRAMS

Program may satisfy needs but may harm computer



- What does it *really* do?
- **LUB tests**
  - **Learn** – What does the program do to your computer?
  - **Understand** – Can you return it and completely remove it?
  - **Buy** – Purchase/download from reputable source?
- **Level of effort: high**



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LECTURE 10